

UNIT - IV

MATERIALS & MANUFACTURING PROCESSES

LECTURE NO. – 25

Introduction to various engineering materials

Materials are an important aspect of engineering design and analysis. The importance of materials science and engineering can be noted from the fact that historical ages have been named after materials. In the customer driven competitive business environment, the product quality is of paramount importance. The product quality has been found to be influenced by the engineering design, type of materials selected and the processing technology employed.

Therefore, the importance of materials and their processing techniques cannot be undervalued in today's world. Materials form the stuff of any engineering application or product. It has been found that the engineers do not give adequate attention to this important subject. Moreover, it has not been adequately represented in the course curriculum of various universities. Therefore, it becomes imperative to highlight the importance of engineering materials for all engineers related to the various aspects of engineering applications.

There is a wide variety of materials available which have shown their potential in various engineering fields ranging from aerospace to house hold applications. The materials are usually selected after considering their characteristics, specific application areas, advantages and limitations. The challenge for designers is to select an optimal material suitable for the specific design requirements. The stringent design requirements generally lead to development of new materials to meet the specific operating conditions and environments. The new materials are developed from the conventional materials by either by the intrinsic or the extrinsic modification.

In intrinsic modification, minor alloying or heat treatment is carried out. In extrinsic modification, external reinforcements are added to the parent material to alter its properties in order to meet the specific design requirements. The composite materials

represent an example of the extrinsic modification. The modification is usually done to improve the properties of the existing materials. As the new materials are conceptualized and realized in the laboratories, the hunt for their commercialization begins.

The engineers are then entrusted with the task of finding suitable techniques which would lead to high quality cost-effective processing of these materials. In order to achieve this objective, it is imperative for all engineers to have a fundamental understanding of the existing materials and their processing techniques. It has been found that there are adequate of courses in the curriculum of various universities where the processing techniques for metals are dealt in detail. The processing of non-metals is usually not covered as a core subject at the under-graduate level and therefore the engineers do not have a fundamental understanding about the processing of important non-metals such as plastics and ceramics. The course has been designed to study the basic nature of different non-metals and the manufacturing processes associated thereof. The various non-metals covered in the course include glasses, ceramics, plastics and different types of composite materials.

Classification and Selection of Materials:

The first module deals with the classification of the engineering materials and their processing techniques. The engineering materials can broadly be classified as:

- a) Ferrous Metals
- b) Non-ferrous Metals (aluminium, magnesium, copper, nickel, titanium)
- c) Plastics (thermoplastics, thermosets)
- d) Ceramics and Diamond
- e) Composite Materials
- f) Nano-materials

The engineering materials are often primarily selected based on their mechanical, physical, chemical and manufacturing properties. The secondary points to be considered are the cost and availability, appearance, service life and recyclability. In the present day, strict environmental legislations have forced the designers and engineers to select the environment friendly materials which have minimum environmental impact. Materials are often selected for their mechanical, physical, chemical and manufacturing

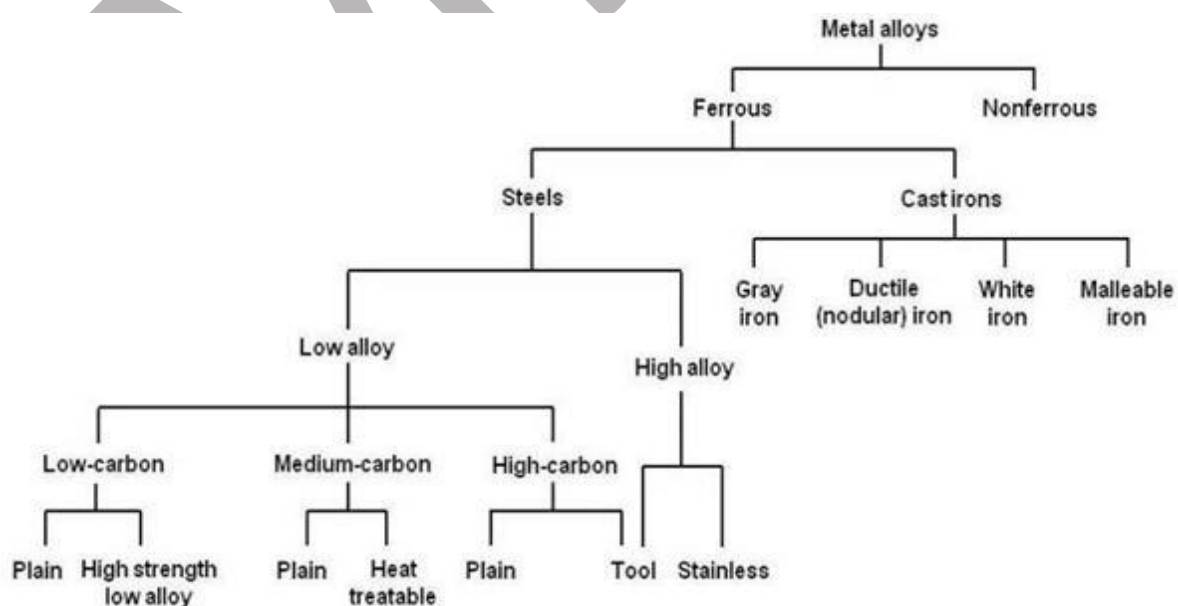
properties for ensuring proper functioning under desired conditions. When several materials are suitable for a particular design requirement, the cheapest one is usually selected.

Classification of Processing Techniques

The basic aim of processing is to produce the products of the required quality at a reasonable cost. The basic processes can be broadly classified as:

- a) Primary Forming Processes
- b) Deformative Processes
- c) Material Removal Processes
- d) Joining Processes
- e) Finishing Processes

Most of the engineering materials are processed either individually or in combination by the above mentioned processes. The processes can further be classified as conventional and advanced processes. The specific application area of each will depend on the design requirements and the ability with which a material renders itself to various processing techniques. The selection of a processing technique for any engineering material would broadly depend on the properties (mechanical, physical, chemical) of the material and the required number of parts to be processed.



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LECTURE NO. – 26

CLASSIFICATION OF METALS AND ALLOYS

FERROUS MATERIALS

Steels

- Steels - alloys of iron-carbon. May contain other alloying elements.
- Several grades are available.
- Low Alloy (11 wt% Cr) –Tool Steel

Low Carbon Steel

- Plain carbon steels
 - very low content of alloying elements and small amounts of Mn.
- Most abundant grade of steel is low carbon steel
 - greatest quantity produced; least expensive.
- Not responsive to heat treatment; cold working needed to improve the strength.
- Good weldability and machinability
- High Strength, Low Alloy (HSLA) steels - alloying elements (like Cu, V, Ni and Mo) up to 10 wt %; have higher strengths and may be heat treated.

Medium Carbon Steel

- Carbon content in the range of 0.3 – 0.6%.
- Can be heat treated - austenitizing, quenching and then tempering.
- Most often used in tempered condition tempered martensite
- Medium carbon steels have low hardenability

- Addition of Cr, Ni, Mo improves the heat treating capacity
- Heat treated alloys are stronger but have lower ductility
- Typical applications – Railway wheels and tracks, gears, crankshafts.

High Carbon Steel

- Carbon content 0.6 – 1.4%
- High C content provides high hardness and strength. Hardest and least ductile.
- Used in hardened and tempered condition
- Strong carbide formers like Cr, V, W are added as alloying elements to form carbides of these metals.
- Used as tool and die steels owing to the high hardness and wear resistance property.

Effects of Alloying Elements on Steel

- **Manganese** – strength and hardness; decreases ductility and weldability; effects hardenability of steel.
- **Phosphorus** – increases strength and hardness and decreases ductility and notch impact toughness of steel.
- **Sulfur** - decreases ductility and notch impact toughness Weldability decreases. Found in the form of sulfide inclusions.
- **Silicon** – one of the principal deoxidizers used in steel making. In low-carbon steels, silicon is generally detrimental to surface quality.
- **Copper** – detrimental to hot-working steels; beneficial to corrosion resistance (Cu>0.20%)
- **Nickel**- ferrite strengthener; increases the hardenability and impact strength of steels.
- **Molybdenum** - Increases the hardenability; enhances the creep resistance of low-alloy steels

Stainless steel

A group of steels that contain at least 11% Cr. Exhibits extraordinary corrosion resistance due to formation of a very thin layer of Cr_2O_3 on the surface.

Categories of stainless steels:

- Ferritic Stainless Steels – Composed of α ferrite (BCC)
- Martensitic Stainless Steels – Can be heat treated.
- Austenitic Stainless Steels – Austenite (γ) phase field is extended to room temperature. Most corrosion resistant.
- Precipitation-Hardening (PH) Stainless Steels – Ultra high strength due to precipitation hardening.
- Duplex Stainless Steels – Ferrite + Austenite

Cast Iron:

- Carbon 2.1- 4.5 wt% and Si (normally 1-3 wt%).
- Lower melting point (about 300 °C lower than pure iron) due to presence of eutectic point at 1153 °C and 4.2 wt% C.
- Low shrinkage and good fluidity and casting ability.
- Types of cast iron: grey, white, nodular, malleable and compacted graphite.

Grey Cast Iron:

- Grey cast iron contains graphite in the form of flakes. Named after its grey fractured surface. C: 3.0 – 4.0 wt%, Si: 1.0 – 3.0 %
- Microstructure: graphite flakes in a ferrite or pearlite matrix
- Weak & brittle in tension (the graphite flake tips act as stress concentration sites). Stronger in compression,
- Excellent damping capacity, wear resistance.
- Microstructure modification by varying silicon content and cooling rate
- Casting shrinkage is low

Nodular or Ductile Iron:

- Addition of Mg and/or Cerium to grey iron converts the graphite flakes to nodules. Normally a pearlite matrix.
- Castings are stronger and much more ductile than grey iron as the stress concentration points existing at the flake tips are eliminated.

White Cast Iron:

- White cast iron – C: 2.5 – 3 wt.%, Si: 0.5 – 1.5%. Most of the carbon is in the form of cementite. Named after its white fracture surface.
- Results from faster cooling. Contains pearlite + cementite, not graphite. Thickness variation may result in non-uniform microstructure from variable cooling
- Very hard and brittle
- Used as intermediate to produce malleable cast iron.

Malleable Cast Iron:

- Carbon: 2.3 – 2.7 wt%, Si: 1.0 – 1.75 % Obtained by heat treating white iron for a prolonged period that causes decomposition of cementite into graphite.
- Heat treatment : Two stages – Isothermal holding at 950 °C and then holding at 720 °C. graphite forms in the form of rosettes in a ferrite or pearlite matrix.
- Reasonable strength and improved ductility (malleable)

Compact Graphite Iron (CGI)

- CGI graphite occurs as blunt flakes or with a worm-like shape (vermicular). Carbon: 3.1 – 4.0 wt%, Silicon: 1.7 – 3.0 wt %. Microstructure and properties are between gray and ductile iron.
- Alloying addition may be needed to minimize the sharp edges and formation of spheroidal graphite. Matrix varies with alloy additions or heat treatment.
- As castable as grey iron, but has a higher tensile strength and some ductility.
- Relatively high thermal conductivity, good resistance to thermal shock, lower oxidation at elevated temperatures.

Applications of Cast iron:

Cast irons are used in wide variety of application owing to the properties like good fluidity, ease of casting, low shrinkage, excellent machinability, wear resistance and damping capacity.

Applications

- Car parts – cylinder heads, blocks and gearbox cases.
- Pipes, lids (manhole lids)
- Foundation for big machines (good damping property)
- Bridges, buildings
- Cook wares – Excellent heat retention

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Non - Ferrous Metals

Copper:

- Copper is one of the earliest metals discovered by man.
- The boilers on early steamboats were made from copper.
- The copper tubing used in water plumbing in Pyramids was found in serviceable condition after more than 5,000 years.
- Cu is a ductile metal. Pure Cu is soft and malleable, difficult to machine.
- Very high electrical conductivity – second only to silver.
- Copper is refined to high purity for many electrical applications.
- Excellent thermal conductivity – Copper cookware most highly regarded – fast and uniform heating.
- Electrical and construction industries are the largest users of Cu.

Copper Alloys:

- Brasses and Bronzes are most commonly used alloys of Cu. Brass is an alloy with Zn.
- Bronzes contain Tin, Aluminum, Silicon or Beryllium.
- Other copper alloy families include copper-nickels and nickel silvers. More than 400 copper- base alloys are recognized.

Family of Cu Alloys		
Alloy	Alloying element	UNS numbers
Brass	Zinc (Zn)	C1xxxx–C4xxxx, C66400–C69800
Phosphor bronze	Tin (Sn)	C5xxxx
Aluminium bronzes	Aluminium (Al)	C60600–C64200
Silicon bronzes	Silicon (Si)	C64700–C66100
Copper nickel, nickel silvers	Nickel (Ni)	C7xxxx

Brass:

- Brass is the most common alloy of Cu – It's an alloy with Zn
- Brass has higher ductility than copper or zinc.
- Easy to cast - Relatively low melting point and high fluidity
- Properties can be tailored by varying Zn content.
- Some of the common brasses are yellow, naval and cartridge.
- Brass is frequently used to make musical instruments (good ductility and acoustic properties).

Bronze

- Copper alloys containing tin, lead, aluminum, silicon and nickel are classified as bronzes.
- Cu-Sn Bronze is one of the earliest alloy to be discovered as Cu ores invariably contain Sn.
- Stronger than brasses with good corrosion and tensile properties; can be cast, hot worked and cold worked.
- Wide range of applications: ancient Chinese cast artefacts, skateboard ball bearings, surgical and dental instruments.

Beryllium copper:

- Cu-Be alloys are heat treatable. Max solubility of Be in Cu is 2.7% at 866 °C. Decreasing solubility at lower temp. imparts precipitation hardening ability.
- Cast alloys - higher Be. Wrought alloys – lower Be and some Co
- Cu-Be is ductile, weldable and machinable. Also resistant to non-oxidizing acids (HCl or H₂CO₃), abrasive wear and galling.
- Thermal conductivity is between steels and aluminum.

Applications

Used in springs, load cells and other parts subjected to repeated loading. Low-current contacts for batteries and electrical connectors. Cast alloys are used in injection molds. Other applications include jet aircraft landing gear bearings and bushings and percussion instruments.

Aluminum

- Aluminum is a light metal, it is easily machinable; has wide variety of surface finishes; good electrical and thermal conductivities; highly reflective to heat & light.
- Versatile metal - can be cast, rolled, stamped, drawn, spun, roll-formed, hammered, extruded and forged into many shapes.
- Aluminum can be riveted, welded, brazed, or resin bonded.
- Corrosion resistant - no protective coating needed, however it is often anodized to improve surface finish, appearance.
- Al and its alloys - high strength-to-weight ratio (high specific strength) owing to low density.
- Such materials are widely used in aerospace and automotive applications where weight savings are needed for better fuel efficiency and performance.
- Al-Li alloys are lightest among all Al alloys and find wide applications in the aerospace industry.

Titanium

- Pure titanium melts at 1670 °C and has a low density of 4.51 g/cc (40% lighter than steel and 60% heavier than aluminum).
- Titanium has high affinity to oxygen – strong deoxidiser. Can catch fire and cause severe damage
- Ti is stronger than Al - high strength and low weight makes titanium very useful as a structural metal.
- Excellent corrosion resistance due to a presence of a protective thin oxide surface film. Can be used as biomaterial.
- Can be used in elevated temperature components.
- Limitation of pure Ti is its lower strength. Alloying is done to improve strength.

Applications:

- Aircraft – Body structure, Engine parts
- sporting equipment, chemical processing, desalination, turbine engine parts, valve and pump parts, marine hardware

- Medical implants - prosthetic devices.
- Recently use of Ti in bikes and automotive is increasing Nickel

Nickel:

- Nickel is a high-density, high-strength metal with good ductility and excellent corrosion resistance and high temperature properties.
- Ni has many unique properties including its excellent catalytic property. Nickel Catalyst for Fuel Cells: Nickel cobalt is seen as a low-cost substitute for platinum catalysts.
- Two-thirds of all nickel produced goes into stainless steel production. Also used extensively in electroplating various parts in variety of applications.
- Ni-base super alloys are a unique class of materials having exceptionally good high temperature strength, creep and oxidation resistance. Used in many high temperature applications like turbine engines.

Magnesium:

- Magnesium - Lightest among commonly used metals (Density = 1.7 g/cm³). Melting point is 650° C and it has HCP structure.
- It is very reactive and readily combustible in air. Can be used as igniter or fire starter.
- Thermal conductivity is less than Al while their CTE is almost same.
- Pure Mg has adequate atmospheric resistance and moderate strength.
- Properties of Mg can be improved substantially by alloying.
- Favourable atomic size - Can be alloyed with many elements. Most widely used alloying elements are Al, Zn, Mn and Zr.
- Mg Alloys – Cast, Wrought
- Wrought alloys are available in rod, bar, sheet, plate, forgings and extrusions.

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LECTURE NO - 28

Introduction and Classification to Manufacturing Process:

Manufacturing:

Manufacturing is the production of products for use or sale using labor and machines, tools, chemical or biological processing or formulation and is the essence of secondary industry.

Manufacturing Process:

Manufacturing engineering or manufacturing process are the steps through which raw materials are transformed into a final product. The manufacturing process begins with the product design, and materials specification from which the product is made. These materials are then modified through manufacturing processes to become the required part.

Classification of Manufacturing Process:

The manufacturing processes can be classified as:

1. Forming Processes
2. Moulding Processes
3. Machining Processes
4. Assembly Processes
5. Finishing Processes.

1. Forming Processes

In the metal industry, some of the primary forming operations may take place such as the rolling of basic shapes in steel, aluminium etc. Some of the common shapes so obtained from these processes are bars, sheets, billets, I-beams etc. Which are standard shapes. These shapes can be used for further processing. Other forming processes may be drop-forging, stamping, extrusion, press work, punching, drawing etc.

2. Moulding Processes

Some products require moulding processes such as sand casting, die-casting etc. to get basic shape or form which may or may not require further processing. The selection of the particular process will depend upon to size of the job, quantity to be produced, accuracy, and complexity desired and economy.

3. Machining Processes

Metal machining is accomplished through basic machine tool processes which involve the generation of cylindrical surfaces, flat surfaces, complex curves and holes. The machine tools selected to accomplish this task depend on the size and shape of the part to be machined, the quality of finish required and production rate required.

The examples of such processes are: turning, shaping, drilling, boring, grinding etc. In these machining operations, metal is removed from the part in the form of small chips by the cutting action of tool. The cutting action is accomplished by either rotating or reciprocating action of the tool relating to the part.

4. Assembly Processes

These processes assemble the parts and materials using welding, riveting, soldering, brazing, mechanical fastening and adhesive joining etc.

5. Finishing Processes

These processes are carried out for the aesthetic aspects, to achieve accuracy, surface finish or to increase life of the product. Such processes include cleaning, blasting, deburring, puffing, honing, lapping, polishing, painting etc.

Introduction and classification of forging processes

Bulk deformation processes involve shaping of materials to finished products which have small surface area to thickness or surface area to volume ratio. Sheet metal forming produces parts having large surface area to thickness ratio. In sheet metal forming thickness variations are not desirable. Examples for sheet metal forming are: beverage cans, automobile body etc. Bulk forming processes may be primary processes such as rolling of

ingot to blooms or billets, in which the cast metal is formed into semi-finished raw material. In secondary forming, the raw materials, such as blooms, billets are converted into finished parts such as gears, wheels, spanners etc.

Forging is a deformation processing of materials through compressive stress. It is carried out either hot or cold. Hot forging is done at temperatures above recrystallization temperatures, typically $0.6 T_m$, or above, where T_m is melting temperature.

Types of forging

In forging the material is deformed applying either impact load or gradual load. Based on the type of loading, forging is classified as hammer forging or press forging. Hammer forging involves impact load, while press forging involves gradual loads. Based on the nature of material flow and constraint on flow by the die/punch, forging is classified as open die forging, impression die forging and flashless forging.

Classification on the basis of temperature of work piece

- Hot forging – forging at high temperature
- Cold forging- forging at room temperature

Classification on the basis of operation of work piece

- A. Hand forging
- B. Drop forging
- C. Press forging

A. HAND FORGING

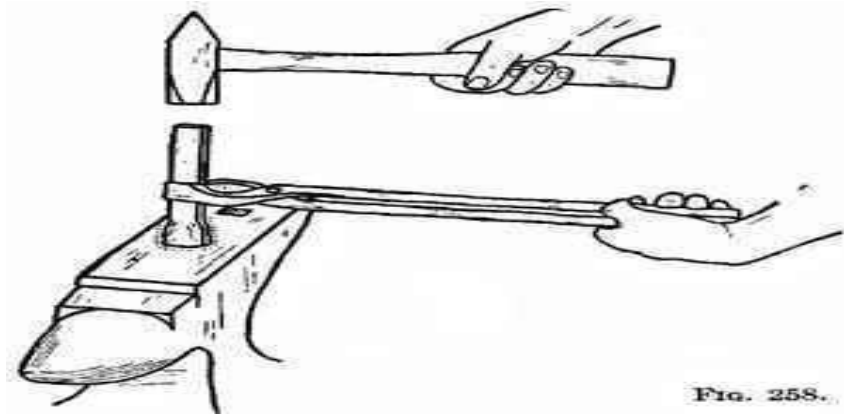
- Traditional forging operation carried out by blacksmith in a section of workshop called smithy
- Hand tools are used for forging (e.g. Hammer, chisel...etc)
- Not suitable for mass production

Hand forging operations

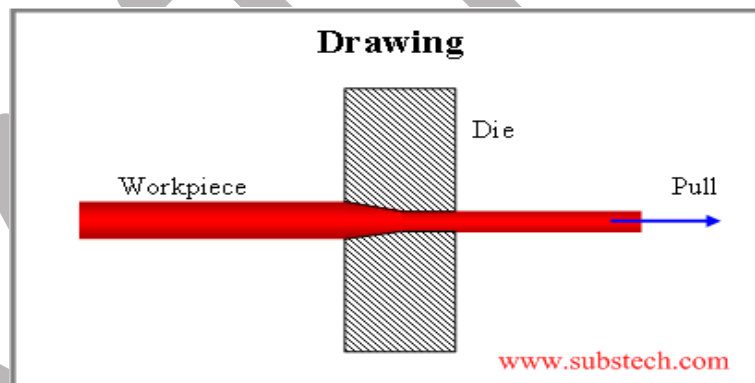
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|--------------|--------------------------------|-----------------|-------------|
| a) Upsetting | b) Drawing down (necking down) | c) Setting down | d) Bending |
| e) Welding | f) Cutting | g) Swaging | h) Drifting |
| | i) Fullering | j) Edging | |

a. Upsetting

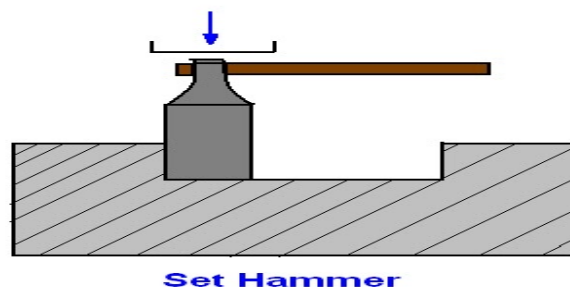
- Process of increasing the cross sectional area of a bar at any desired portion at the expense of length of the bar.
- Portion to be upset is heated and then hammered axially

**b. Drawing Down**

Process of reducing the cross section of a bar by increasing its length.

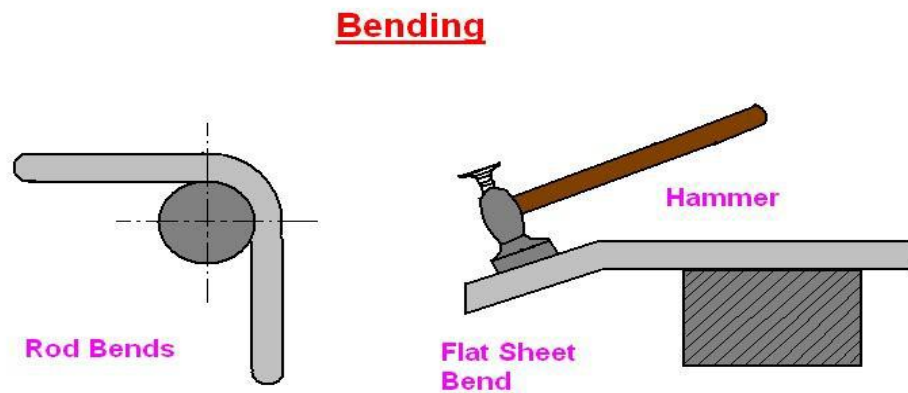
**c. Setting Down**

Local thinning down operation using a set of hammer.

Setting Down

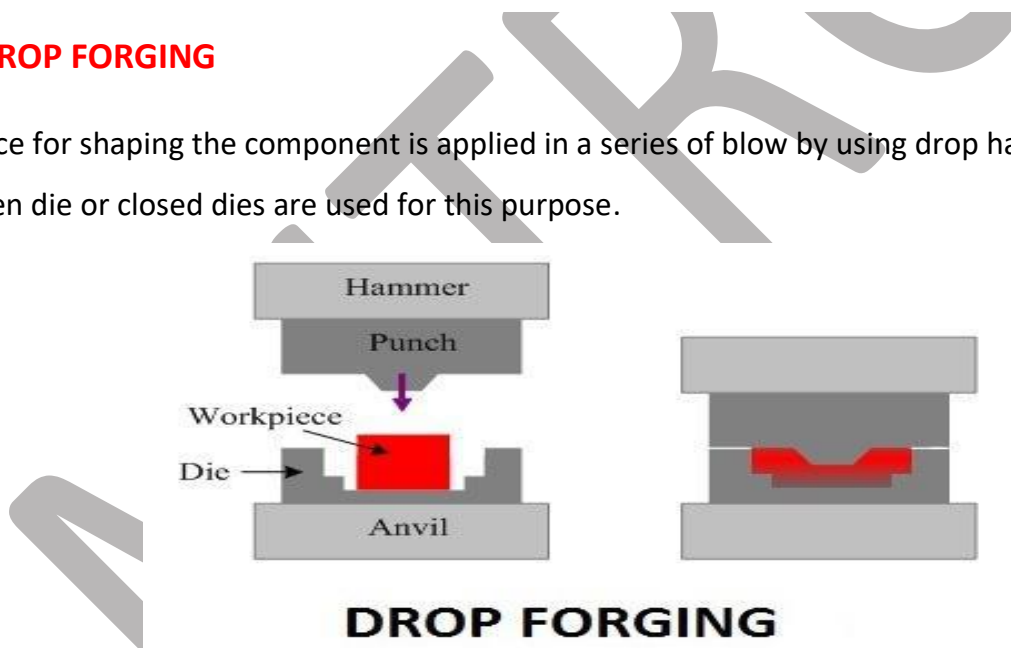
d. Bending

Bars and rods are bent to form rings, hooks...etc



B. DROP FORGING

- Force for shaping the component is applied in a series of blow by using drop hammers
- Open die or closed dies are used for this purpose.



C. PRESS FORGING

- Process is similar to drop forging but for the method of application of force
- In this case the force is applied by a continuous squeezing operation by means of a hydraulic press.

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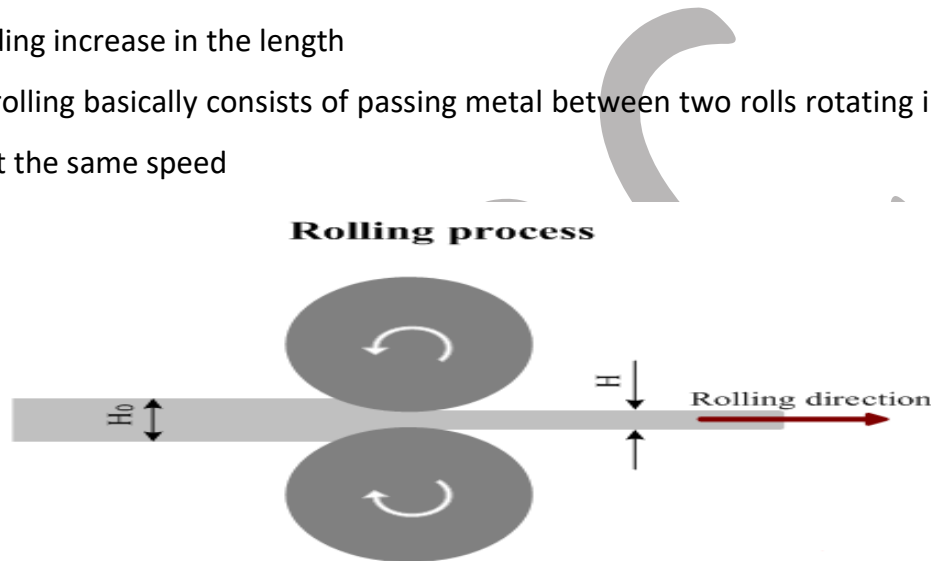
MATERIALS & MANUFACTURING PROCESSES

LECTURE NO. - 29

ROLLING

Process of plastically deforming metal by passing it between rolls

- Cylindrical rolls are used to reduce the cross-sectional area of a bar or plate with a corresponding increase in the length
- Process of rolling basically consists of passing metal between two rolls rotating in opposite direction at the same speed



Rolling are two types:-

- Hot rolling
- Cold rolling

1. Hot Rolling

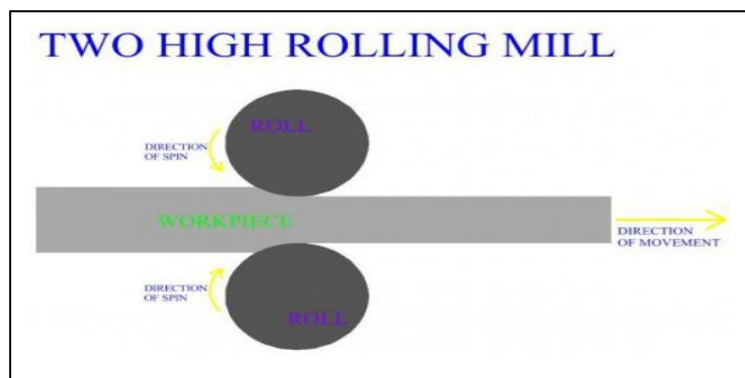
Process in which metal is fed to the rolls after being heated **above the recrystallization temperature**.

2. Cold Rolling

In cold rolling, metal is fed to the rolls when it is **below the recrystallization temperature**

Types of rolling mills

- Two high mill
- Three high mill
- Four high mill
- Cluster mill
- Tandem mill



EXTRUSION PROCESS

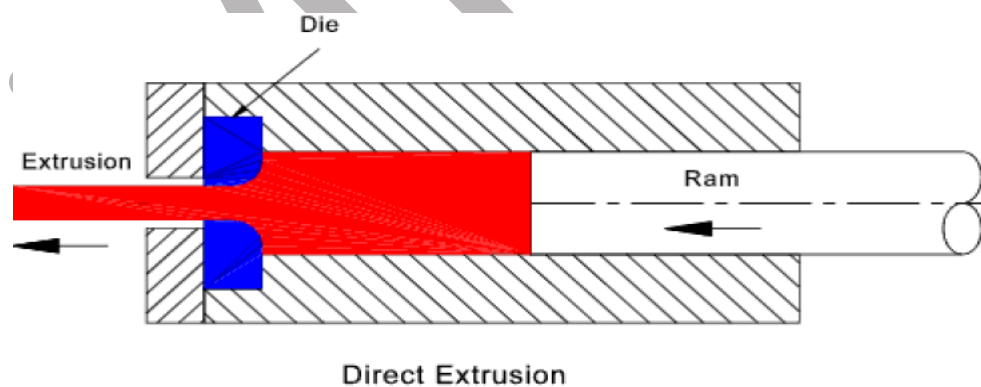
- Process of forcing a metal enclosed in a container to flow through the opening of a die
- Metal is subjected to plastic deformation
- Metal undergoes reduction and elongation during extrusion
- Used for manufacture rods, tubes, circular, rectangular, hexagonal and other shapes both in hollow and solid form

Types of Extrusion

- a) Direct Extrusion
- b) Indirect Extrusion
- c) Cold Extrusion/ Impact Extrusion

A. Direct Extrusion

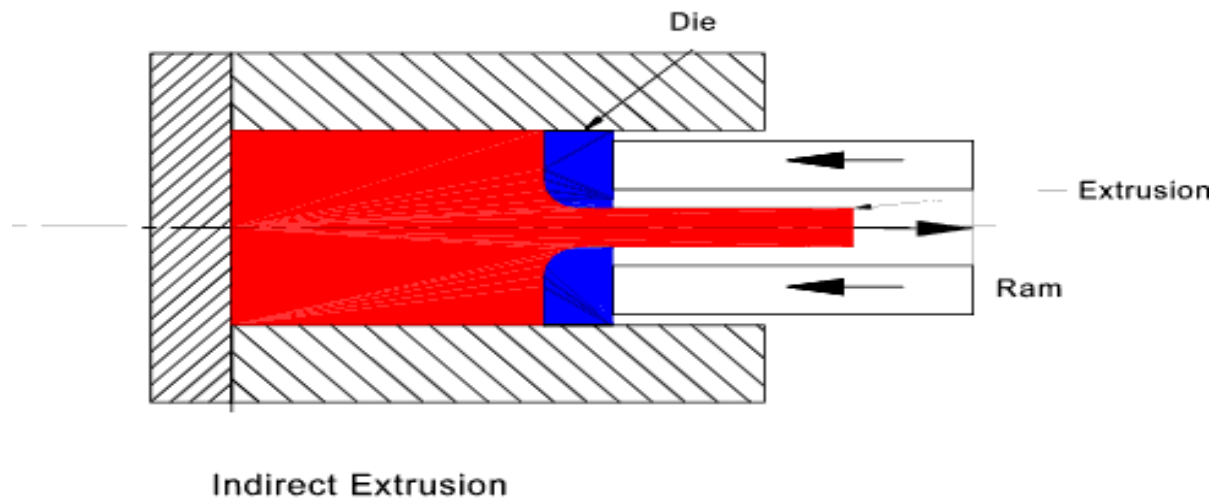
- Also called forward extrusion
- *Flow of metal through the die is in the same direction as the movement of ram*
- Hot billet (work piece) is used



B. Indirect Extrusion

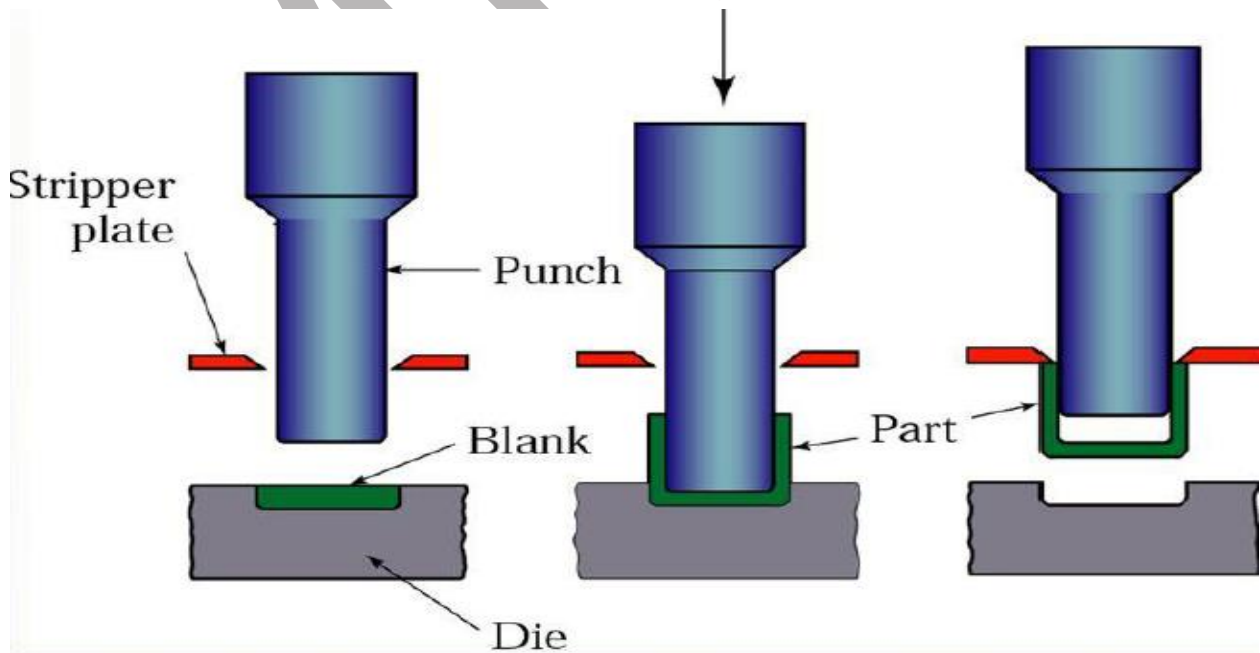
- Also called backward extrusion
- Flow of metal through the die is in the opposite direction as the movement of ram
- Hot billet (work piece) is used
- Ram used is hollow

- Billet remains stationary while die is pushed into the billet by the hollow ram
- Less force is required as compared to direct extrusion



C. Cold/Impact Extrusion

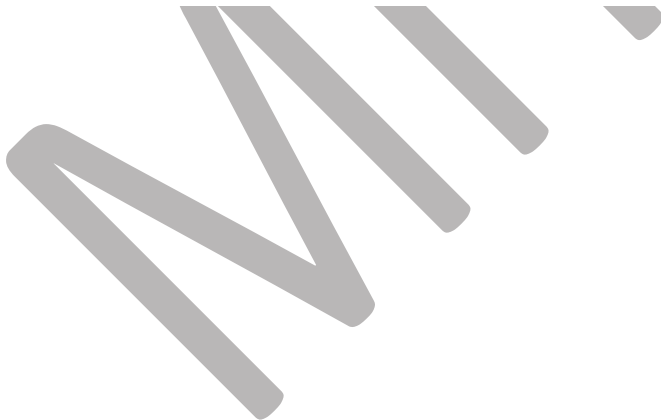
- Carried out at higher velocity
- Unheated metal is placed in the die cavity
- Punch is forced into the die cavity causing the metal to flow upwards through the gap between punch and die
- Tooth pastes tubes are made by this process



DRAWING

Drawing is a metalworking process which uses tensile forces to stretch metal or glass. as the metal is drawn (pulled), it stretches thinner, into a desired shape and thickness. Drawing is classified in two types: sheet metal drawing and wire, bar, and tube drawing. The specific definition for sheet metal drawing is that it involves plastic deformation over a curved axis. For wire, bar, and tube drawing the starting stock is drawn through a die to reduce its diameter and increase its length. drawing is usually done at room temperature, thus classified a cold working process, however it may be performed at elevated temperatures to hot work large wires, rods or hollow sections in order to reduce forces.

Drawing differs from rolling in that the pressure of drawing is not transmitted through the turning action of the mill but instead depends on force applied locally near the area of compression. This means the amount of possible drawing force is limited by the tensile strength of the material, a fact that is particularly evident when drawing thin wires.



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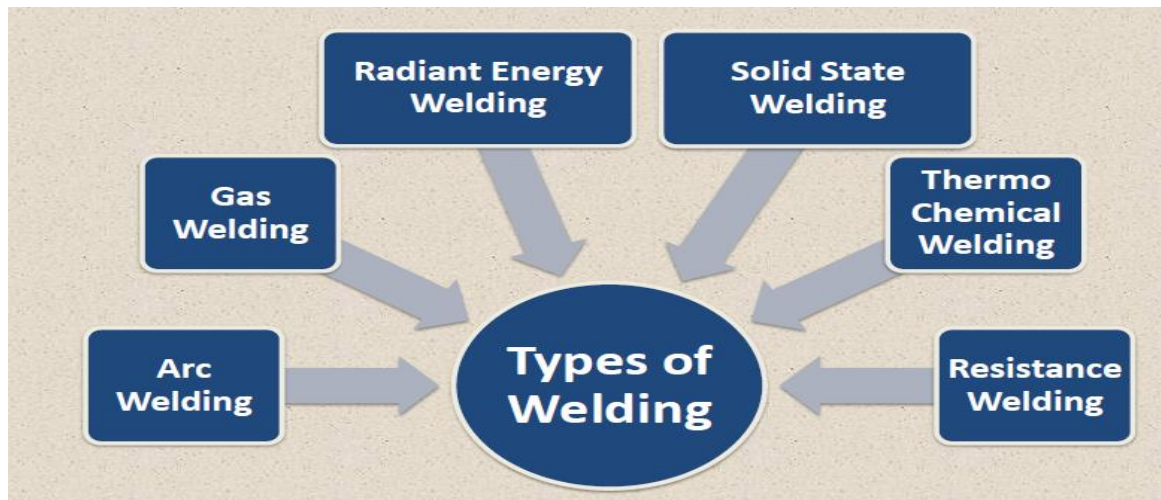
JOINING PROCESS

Introduction to various types of Welding

Process of joining similar or dissimilar metals by the application of heat, with or without the application of pressure and with or without the addition of filler material. Welding is a [fabrication](#) or [sculptural process](#) that joins materials, usually [metals](#) or [thermoplastics](#), by causing [fusion](#), which is distinct from lower temperature metal-joining techniques such as [brazing](#) and [soldering](#), which do not [melt](#) the base metal. In addition to melting the base metal, a filler material is typically added to the joint to form a pool of molten material (the [weld pool](#)) that cools to form a joint that is usually stronger than the base material. [Pressure](#) may also be used in conjunction with [heat](#), or by itself, to produce a weld. Welding also requires a form of shield to protect the filler metals or melted metals from being contaminated or [oxidized](#).

Welding are Two types

- Plastic welding
- Fusion welding
- Plastic welding:** metals to be joined are to be heated to the plastic state and then forced together by external pressure without the addition of filler material. Eg. forge welding, resistance welding...etc
- Fusion welding:** no pressure is involved but a very high temperature is produced in or near the joint. The metal at the joint is heated to the molten state and allowed to solidify. A filler material may be used during the welding process. Eg. Oxy-acetylene welding, carbon arc welding...etc.



1. GAS WELDING

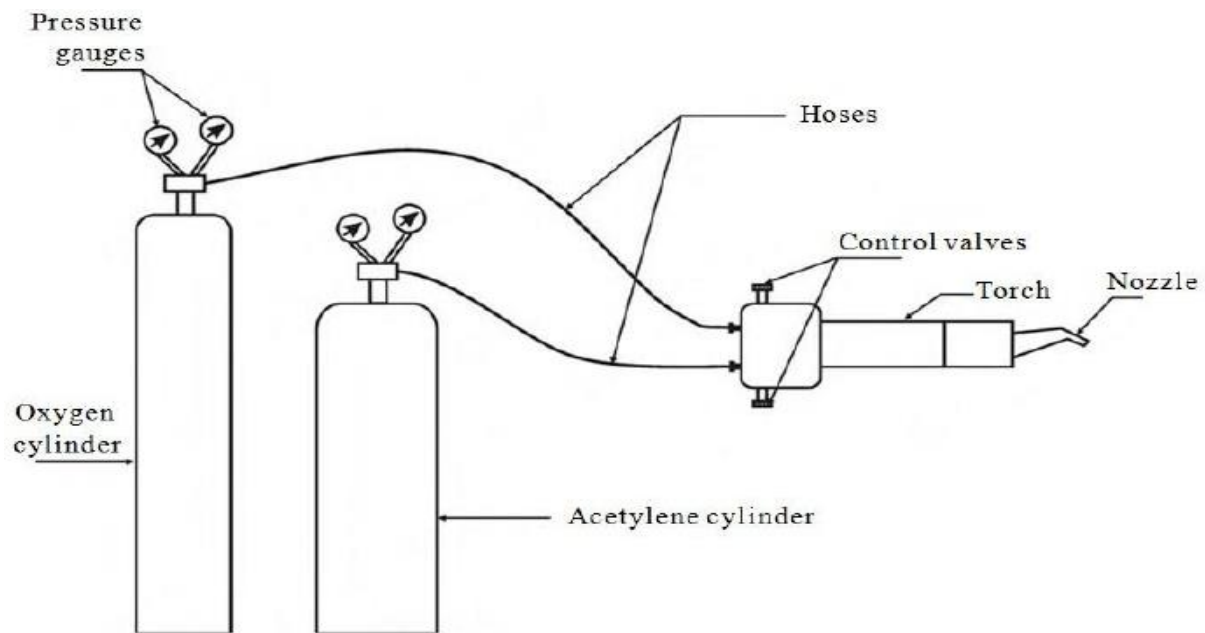
- Welding process that uses a fuel gas combined with oxygen to produce a flame
- Fusion welding process
- It joins metals using the heat of combustion of oxygen/air and fuel gas (acetylene, hydrogen, propane or butane)

Gas welding are three types

- Oxy-Acetylene Welding
- Air-Acetylene Welding
- Oxy-Hydrogen Welding

GAS WELDING: OXY-ACETYLENE WELDING

- It joins the metals using the heat of combustion of oxygen and fuel gas (acetylene)
- Acetylene gas is mixed with oxygen in correct proportions in the welding torch and ignited, the flame resulting at the tip of the torch is sufficiently hot (3200°C) to melt and join the metal
- Volume of gas passing through the torch is controlled by two control valves provided in the torch
- The gases are mixed in the mixer part of the torch, before delivering to the torch tip
- Filler metals are additional metal added to the weld
- Flux is added to remove impurities and oxides (flux reacts with oxides and a slag is formed) formed during welding operation



Gas Welding

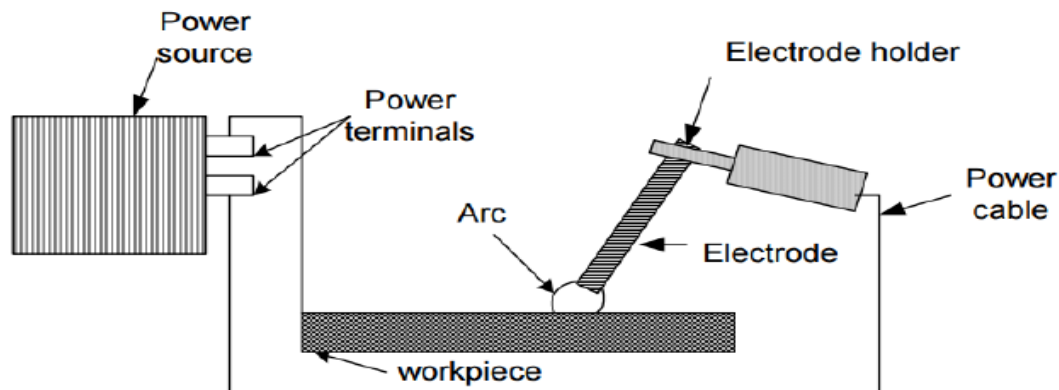
2. ARC WELDING

- Method of fusion welding in which the metals at the joint is heated to molten state by an electric arc
- Arc column is generated between an anode (electrode) and the cathode (metal to be joined)
- Temperature of the arc is about 6000°C to 7000°C.

ARC WELDING- SHIELDED METAL ARC WELDING

- The heat is generated by an electric arc between base metal and a consumable electrode.
- In this process electrode movement is manually controlled hence it is termed as manual metal arc welding
- In shielded metal arc welding, the protection to the weld pool is provided by covering of
 - a) Slag formed over the surface of weld pool/metal and
 - b) Inactive gases generated through thermal decomposition of flux/coating materials on the electrode
- This process can use both AC and DC. The constant current DC power source is invariably used with all types of electrode irrespective of base metal (ferrous and non-ferrous)
- AC can be unsuitable for certain types of electrodes and base materials

- **SHIELDED METAL ARC WELDING** normally uses constant current type of power source with welding current 50-600A and voltage 20-80V
- Welding current (A) is generally selected in range of 40-60 times of electrode diameter (mm)



Schematic diagram showing various elements of SMA welding system

ELECTRODES

- **Non- consumable electrodes:** carbon, graphite or tungsten (they are not consumed during welding operation)
- **Consumable electrodes:** steel (consumed during welding operation), electrode may be bare (uncoated) or flux coated.

3. RESISTANCE WELDING

- Parts to be joined are heated to a plastic state over a limited area by their resistance to flow of electric current (resistance heating) and then by mechanically pressing together.
- Pressure is applied continuously till the weld cools down
- It is generally used for joining thin plates and structures

Types of Resistance Welding

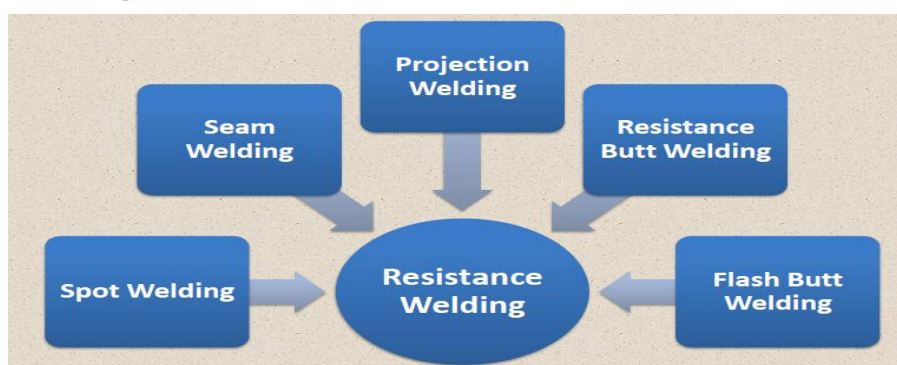


Fig. Types of Resistance Welding

ADVANTAGES, DISADVANTAGES & APPLICATION OF WELDING

ADVANTAGES OF WELDING

The main advantages of welding are as follows:

1. Welding joint is stronger than other joints.
2. Welding joints takes longer time and less costing.
3. The efficiency of welding joint is 100% which is not in other joints.
4. Welding joint remains strong at high temperature.
5. Lesser metal is required in filling the welding joint and time is also saved.
6. Welding can be used in repairing work easily and quickly.

DISADVANTAGES OF WELDING

Disadvantages of welding are as follows:

1. Many welding techniques require high costing.
2. The property of the metals changes at high temperature in welding process.
3. Various types of poisonous gases and smoke are produced in welding processes which is harmful for the health of the welder.
4. Ultraviolet rays produced in welding processes are harmful for the eyes.
5. Skilled labours are required for high level welding process.

APPLICATIONS OF WELDING

The main uses of welding are as follows:

1. For making aircrafts.
2. For making and repairing train compartments.
3. In small and large bridge construction.
4. In automobile industries- making of moped, scooter, car, truck, bus, etc.
5. In tractor industries- making of tractor and trolley.
6. In construction industries- construction of gates, railings, doors and stairs of huge buildings.
7. In construction of dam.
8. Cycle industries- in manufacturing of cycles.
9. Heavy machinery- in manufacturing of heavy machinery like-JCB machine, bulldozer, roller, etc.
10. Repairing work- used in repairing related work.

UNIT - IV

MATERIALS & MANUFACTURING PROCESSES

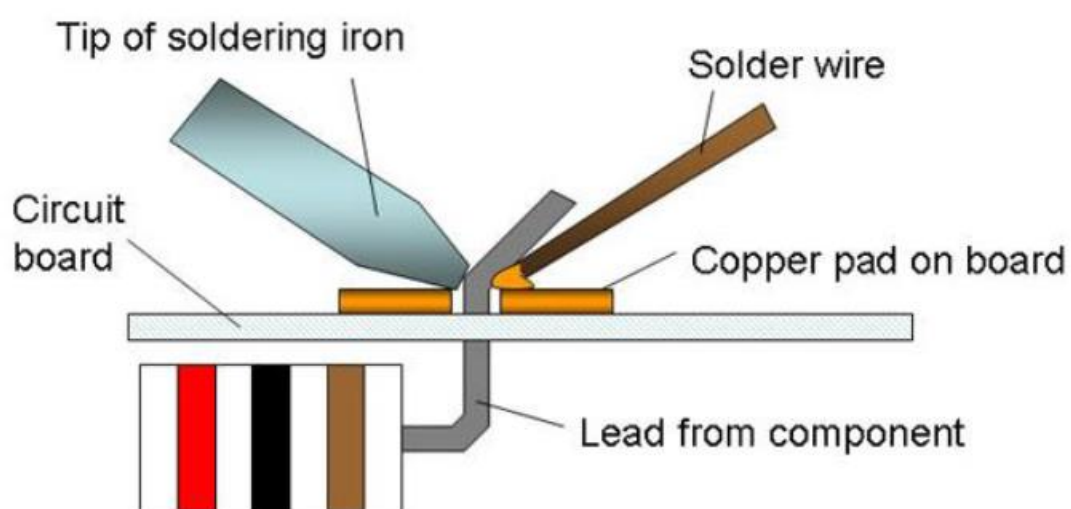
LECTURE NO. - 31

METAL JOINING PROCESSES

SOLDERING

Soldering is a simple and fast means for joining sheet metal, making electrical connections, and sealing seams against leakage. Like welding, soldering uses a filler metal (commonly called solder) to join two metals. However, unlike welding, soldering joins the two metals without heating them to their melting points. In addition, you can also use soldering to join dissimilar metals such as iron, nickel, lead, tin, copper, zinc, aluminium, and many other alloys. 800°F is a key determining temperature. Because solder's melting temperature is below 800°F, it is not classified as a welding or brazing process. Welding and brazing usually take place above 800°F, the one exception being lead welding, which occurs at 621°F. Do not confuse "silver soldering" with soldering. The silver soldering process is a form of brazing because it uses a temperature above 800°F.

- Method of joining two or more metal pieces by means of a fusible alloy or metal called solder (filler alloy), applied in molten state.



- The melting temperature of filler metal is lower than 450°C and also lower than the melting point of the components to be joined.

- No direct melting of the metals being joined.
- During the process, the filler alloy (solder) flows between the two closely adjacent surfaces of the work pieces by capillary action.
- Solder is melted by using a soldering iron, which is heated by electrical resistance.
- Soldering iron tips are made of copper core plated with iron. The copper is used for heat transfer and the iron plating is used for durability

Equipment

Soldering requires very little equipment. Typically, you only need a heat source, a soldering copper or iron, solder, and flux.

Sources of Heat

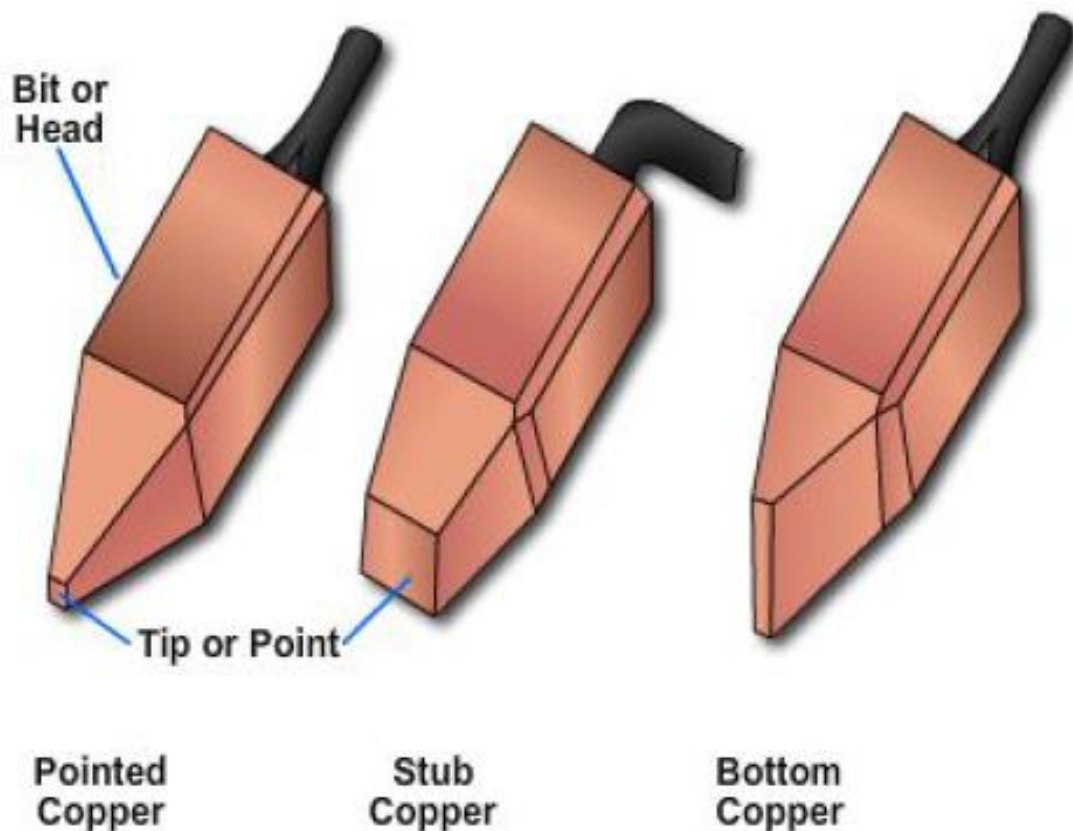
Heat sources can vary according to the available equipment and the method you need to use. Some common sources are welding torches, blowtorches, forges, and furnaces, all of which heat the soldering coppers that secondarily supply the direct heat to the metal surfaces, thus melting the solder. Occasionally you may opt to use a heat source directly to heat the metal, but if or when you do this, you must be careful not to damage the metal or the surrounding material.

Soldering Coppers

Externally heated soldering coppers (soldering irons) consist of a forged copper head, an iron rod, and a handle, usually wood or fiber either screwed or forced on. Other soldering irons are electrically heated. Soldering heads are available in various shapes. Figure below shows three of the more commonly used types.



Examples of soldering coppers (soldering irons)



Common soldering copper heads

Pointed copper — for general soldering work.

Stub copper — for soldering flat seams needing a considerable amount of heat.

Bottom copper — for soldering hard to reach seams, such as those in pails, pans, trays, and similar objects.

Nonelectrical coppers come in pairs so you can use one copper as the other is heating. When coppers are referred to by size designation, they are referred to by weight (in pounds) of the pair, so a reference to a pair of 4-pound coppers means each copper head weighs 2 pounds. Pairs of coppers are usually available in 1-pound, 1 1/2-pound, 3-pound, 4-pound, and 6-pound sizes. As you would expect, because of their differing heat transfer ranges, heavy coppers are designed for soldering heavy gauge metals, while light coppers are for thinner metals, and using the incorrect size of copper usually results in poorly soldered joints from problems caused by not enough steady heat or by overheating.

Electric Soldering Coppers

Electric soldering coppers (usually called soldering irons) use internal heating coils to heat the head, and the heads are removable and interchangeable. Tinning is the same with the exception that the tip usually does not become cherry red as the internal coils have limiting resistors. Forging or reshaping is not necessary since the heads are easily replaced. Electric soldering irons are especially suited for, and usually used for, electrical work or other small jobs (*Figure below*). They do not require auxiliary heating and can be as small as a pencil.



Fig. Typical electric soldering iron with replaceable heads

Advantages & Disadvantages of Soldering

Advantages

The main advantages of soldering are as follows:

1. It can be quickly joined at low temperature.
2. Special expertise or efficiency is not required in soldering.
3. Thin sheets, wires & small parts can be temporarily connected.
4. The costing of this joint is less.

Disadvantages

The main disadvantages caused by soldering are as follows:

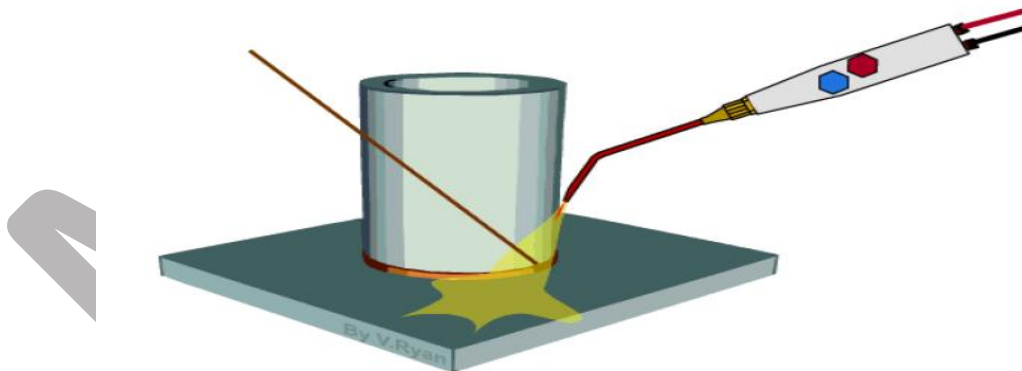
1. This is a weak joint.
2. It detaches on little heating.
3. It cannot be done on thick sheets & parts.

UNIT - IV
MATERIALS & MANUFACTURING PROCESSES
LECTURE NO. - 32

METAL JOINING PROCESSES

BRAZING

- Metal pieces are joined by heating the closely placed parts and then filler alloy called spelter applied in the molten state which upon solidification produces the desired joint.
- Melting temperature of filler metal is more than 450°C but lower than the melting temperature of the components to be joined.
- No direct melting of the metals being joined.
- Brazing gives a much stronger joint compared to soldering.
- During the process, the filler alloy (spelter) flows between the two closely adjacent surfaces of the work pieces by capillary action.
- Torch brazing is the most versatile method (Oxygen-Acetylene)



Advantages of brazing

- Brazing can be used to join a large variety of dissimilar metals.
- Pieces of different thickness can be easily joined by brazing
- Thin-walled tubes & light gauge sheet metal assemblies not joinable by welding can be joined by brazing.
- Complex & multi-component assemblies can be economically fabricated with the help of brazing.
- Inaccessible joint areas which could not be welded by gas metal or gas tungsten arc spot or seam welding can be formed by brazing.

Brazing fluxes

Characteristics of a good flux include,

- (1) low melting temperature,
- (2) low viscosity so that it can be displaced by the filler metal,
- (3) facilitates wetting, and
- (4) protects the joint until solidification of the filler metal.

On the basis of method of heating, various brazing process are:

Torch brazing

- Flux is applied to the part surfaces and a torch is used to direct a flame against the work in the vicinity of the joint.
- After the work part joint areas have been heated to a suitable temperature, filler wire is added to the joint, usually in wire or rod form.
- Fuels used in torch brazing include acetylene, propane, and other gases, with air or oxygen.
- Torch brazing is often performed manually, and skilled workers must be employed.

Other Types of Brazing:

- 1) Furnace brazing
- 2) Induction brazing
- 3) Resistance brazing
- 4) Dip brazing
- 5) Infrared brazing

Brazing Procedure

Brazing procedure is completed in the following steps as per the serial number:

1. Cleaning of the surface to be brazed.
2. Applying flux on the job parts.
3. Heating the job.
4. Applying filler metal.
5. Cleaning of the final brazed parts.

DIFFERENCE BETWEEN SOLDERING & BRAZING

S. No.	Soldering	Brazing
1.	Its joint is weak.	Its joint is strong.
2.	It has low melting point.	It has high melting point.
3.	Lead and tin alloy is used in it.	Copper-zinc, copper-silver and silver-zinc alloys are used in it.
4.	Its joint cannot bear jerks or vibrations.	Its joint can bear vibrations and jerks.
5.	Light joints are made by it like- electricity wires, TV., tape recorder, thin sheets, etc.	Jobs or parts are connected by it.
6.	Its joint can be easily taken out by heating it.	Its joint cannot be separated easily by heating.
7.	Different flux is used in it as per the metals.	Only borax is used as flux.